

$\Sigma^T X \rightarrow \text{Gaussian } \forall X \in \mathbb{R}^d \quad \Sigma \neq 0.$

$\forall X \in \mathbb{R}^d \rightarrow$

Gaussian Vector is determined by:

$E X = \mu \in \mathbb{R}^d$ & Cov. Matrix Σ .

$$X \sim N_d(\mu, \Sigma).$$

PDF:

$$\frac{1}{\sqrt{(2\pi)^d \det(\Sigma)}} e^{-\frac{1}{2} (X-\mu)^T \Sigma^{-1} (X-\mu)}.$$

$\downarrow$ for 1d)

$$\frac{1}{\sqrt{(2\pi)} \det \Sigma}$$

$$e^{-\frac{1}{2} (X-\mu)^2 \Sigma^{-1} (X-\mu)}$$

For 1x1 matrix, $\Sigma = [\sigma^2]$

$X \in \mathbb{R}, \text{Cov}(X, X) = \sigma^2 = \text{Var}(X)$

$\downarrow$

$$\Sigma^{-1} = \frac{1}{\sigma^2}$$

$$\det \Sigma = \sigma^2$$

Multi-Variate Gaussian R.V.:

$$X = (x^{(1)}, \dots, x^{(d)})^T \rightarrow \text{Gaussian Vector.}$$

If: Any linear combination of it's components, is a univariate Gaussian variable or a constant

A "Gaussian variable" with zero variance,

If: $\alpha^T X$ is univariate Gaussian or constant for any $\alpha \in \mathbb{R}^d$ & $\alpha \neq 0$

$$\mu = E[X] = (E[x^1] \dots E[x^d])^T \rightarrow$$

$\downarrow$
dxd covariance matrix Σ ,

If Σ is invertible,

$$\text{pdf: } f(x) = \frac{1}{\sqrt{(2\pi)^d \det(\Sigma)}} e^{-\frac{1}{2} (x-\mu)^T \Sigma^{-1} (x-\mu)} \quad x \in \mathbb{R}^d.$$

$\downarrow$
+ve, when Σ is invertible.

If $\mu = 0$, Σ is the identity matrix,
 X is called the std. Normal vector,

$\Sigma \rightarrow$ Diagonal Matrix,

Pdf of Univariate Gaussians,
and hence the components are independent.

Example:

$$X = \begin{pmatrix} x^{(1)} \\ x^{(2)} \end{pmatrix}, \quad \Sigma_X = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}, \quad \mu_X = \begin{pmatrix} 0 \\ 2 \end{pmatrix}.$$

$$Y = \alpha^T X, \quad \alpha = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$
$$= (x^{(1)} - x^{(2)}).$$

$$\text{Var}(Y) = ?$$

Soln:

$$\frac{1}{\sqrt{2\pi}} \int e^{-\frac{x^2}{2}} \begin{bmatrix} 5 & -2 \\ -2 & 1 \end{bmatrix}$$

DIAGONALISATION OF THE COVARIANCE MATRIX:

$$\Sigma \rightarrow d \times d.$$

$$\rightarrow \Sigma = \Sigma^T \rightarrow \text{SYMMETRIC.}$$

$\rightarrow \Sigma$ is diagonalizable to a diagonal matrix D via a transformation, $D = U \Sigma U^T$, where U is an orthogonal matrix ($A A^T = A^T A = I \rightarrow \text{ORTHOGONAL}$).



$$\boxed{\Sigma = U^T D U.}$$

$\rightarrow \Sigma$ is +ve semi-definite. , Diagonal entries ≥ 0 .

$\rightarrow \Sigma$ has a unit square-root. $\Sigma^{1/2} \cdot \Sigma^{1/2} = \Sigma$.



$\Sigma^{1/2} \rightarrow \text{unique.}$

$\rightarrow \Sigma \rightarrow d \times d \rightarrow d$ orthonormal eigenvectors
(even if repeated eigen values)

If U is a matrix with rows corresponding to the orthonormal eigen vectors, then the diagonal matrix $D = U \Sigma U^T$ contains the eigen-values of Σ along its diagonal.

$\Rightarrow$ Diagonalization of a symmetric matrix involves finding its eigenvalues & orthonormal eigenvectors.

$\Sigma \rightarrow$ +ve definite, Diagonal Matrix $D = U \Sigma U^T$.

Diagonal entries +ve, it is INVERTIBLE.

Σ^{-1} INVERSE SATISFIES: $\Sigma^{-1/2} \cdot \Sigma^{-1/2} = \Sigma^{-1}$

$\Sigma^{-1/2} \rightarrow$ Inverse of the square-root of Σ .

MULTI-VARIATE CLT:

Convergence in Distribution

→ Multi-variate Gaussian as → Limits of CLT.

→ Average of a random-vector →

↓
vector of the averages.

→ Bunch of vectors → Take their averages.

↓
Average has a
co-ordinate in each of the
dimensions

→ $x_1, \dots, x_n \in \mathbb{R}^d \rightarrow$ Independent copies of
R.V. x , such that $E[x] = \mu$, $\text{Cov}(x) = \Sigma$.

$$\sqrt{n}(\bar{x}_n - \mu) \rightarrow N_d(0, \Sigma)$$

↑
 $E[x] \rightarrow d$ dimensions ($d \times 1$)
 $I_d \rightarrow$ Identity matrix

I_d of $\mathbb{R}^d$

↑
 $I_d =$

$$\begin{bmatrix} 1 & & & \\ & \ddots & & \\ & & \ddots & \\ & & & 1 \end{bmatrix}$$

→ I want to divide the previous equation by $\Sigma^{1/2}$ to get Id.

$$\Sigma^{1/2} \rightarrow \sigma \rightarrow \underline{In - 1-d}$$

$$\sqrt{n} \Sigma^{-1/2} (\bar{X}_n - \mu) \rightarrow N_d(0, Id).$$

↑
(To get this, we need to get Eigen values)

$\Sigma^{-1/2} \rightarrow d \times d$ matrix.

s.t. -

$$\Sigma^{-1/2} \cdot \Sigma^{-1/2} = \Sigma^{-1}$$

we can talk about this

I need to think of this matrix as ≥ 0

(Co-Variance matrices are +ve definite.)

$$\Sigma \geq 0$$

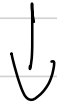
↓
And vice-versa

UNLESS.

UNLESS

Some variances are zero, we relax the sign $\rightarrow$ To allow the matrix $= 0$.

$\downarrow$
In this class we won't.



Look at Wikipedia
for inverse & sq. root

Proof of Multi-Variate CLT:

CONVERGENCE IN HIGHER DIMENSIONS!

$\rightarrow$ Random - vector $\in \mathbb{R}^d$ (Convergence in Distribution)
~~CLT~~
Convergence of each of its components

$\rightarrow$ $T_1, T_2, \dots \rightarrow \mathbb{R}^d$ Convergence in Distribution to $\bar{\tau}$
sequence

if $v^T \cdot T_n \xrightarrow[n \rightarrow \infty]{(d)} v^T T$ for all $v \in \mathbb{R}^d$.

Vector Sequence $T_n, n \geq 1$ convergence in distribution only if it's dot Product $v^T \cdot T_n$

(with any constant vector v)

CONVERGES
IN DISTRIBUTION

(Scalar Random
Variable output)

Projection of vector sequence
onto any line, converges in distribution.

UNIVARIATE CLT IMPLIES

MULTI-VARIATE CLT:

$X_i \stackrel{iid}{\sim} X \rightarrow X_i \in \mathbb{R}^d$. $E[X] = \mu_X$ $\Sigma_X \rightarrow$
(Covariance matrix)
(of 1 Random Vector)

Let $v \in \mathbb{R}^d$ and define $Y_i = v^T X_i$.

$Y_i \rightarrow \text{scalar}$. $E[Y_i] = v^T E[X_i]$ $\sigma_Y^2 = v^T \Sigma_X v$
 $\downarrow$
scalar.

Y_i satisfies the univariate CLT:

$$\sqrt{n} (\bar{Y}_n - v^T \mu_X) \xrightarrow[n \rightarrow \infty]{(d)} N(0, v^T \Sigma_X v).$$

A multi-variate Gaussian R.V. $Z \sim N(0, \Sigma_X)$

For any constant vector $v \in \mathbb{R}^d$, $v^T Z$ is a
UNIVARIATE GAUSSIAN,

variance: $v^T \Sigma_X v$.

Hence

$$v^T Z \sim N(0, v^T \Sigma_X v)$$

Hence $\bar{X}_n$ converges in distribution

$$\begin{aligned} \sqrt{n} (v^T [\bar{X}_n] - v^T \mu_X) &= \sqrt{n} [\bar{Y}_n - v^T \mu_X] \\ &\xrightarrow[n \rightarrow \infty]{(d)} N(0, v^T \Sigma_X v) \\ &= N^T N(0, \Sigma_X). \end{aligned}$$

$$\iff \sqrt{n} (\bar{X}_n - \mu_X) \xrightarrow[n \rightarrow \infty]{(d)} N(0, \Sigma_X)$$

MULTI-VARIATE Δ -METHOD:

→ single $g(\bar{x}_n) - g(\theta) \xrightarrow{(d)} N(0, (g'(\theta))^2 \sigma^2)$.

CLT:

Let T_n be a sequence $\in \mathbb{R}^d$ s.t.

$$\sqrt{n}(T_n - \theta) \xrightarrow[n \rightarrow \infty]{(d)} N_d(0, \Sigma)$$

$\theta \in \mathbb{R}^{(d)} \quad \Sigma \in \mathbb{R}^{d \times d}$

Let $g: \mathbb{R}^d \rightarrow \mathbb{R}^{(k)} \ (k \geq 1)$, continuously differentiable at θ ,

$$\sqrt{n}(g(T_n) - g(\theta)) \xrightarrow[n \rightarrow \infty]{(d)} N_d(0, [\nabla g(\theta)]^T \Sigma \nabla g(\theta))$$

Earlier we had $\mathbb{R}^d \rightarrow \mathbb{R}$.

$$\nabla g(\theta) = \frac{\partial g(\theta)}{\partial \theta}$$

$$\begin{pmatrix} \frac{\partial g_1}{\partial \theta_1} & \dots & \frac{\partial g_1}{\partial \theta_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_k}{\partial \theta_1} & \dots & \frac{\partial g_k}{\partial \theta_d} \end{pmatrix} \in \mathbb{R}^{d \times k}$$

$1 \leq i \leq d$
 $1 \leq j \leq k$

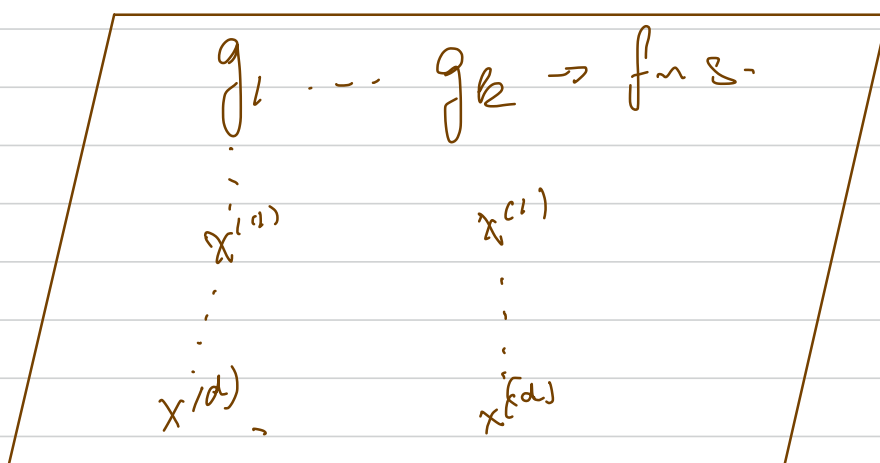
Let $g: \mathbb{R}^d \rightarrow \mathbb{R}$, $g(x^{(1)}, \dots, x^{(d)})$ $\in \mathbb{R}$

$$(\nabla g) = \begin{pmatrix} \frac{\partial}{\partial x_1} g(x_1) \\ \vdots \\ \frac{\partial}{\partial x_d} g(x_d) \end{pmatrix}$$

Now, $g: \mathbb{R}^d \rightarrow \mathbb{R}^k$.

$$g(x) = \begin{pmatrix} g_1(x_1, \dots, x_d) \\ \vdots \\ g_k(x_1, \dots, x_d) \end{pmatrix}$$

one fun. per
co-ordinate.



If I want to talk about derivatives,

I want to talk about derivatives per co-ordinate.

∴

$$\nabla g = \begin{bmatrix} \frac{\partial}{\partial x_1} g_1(x) & \frac{\partial}{\partial x_1} g_2(x) \\ \vdots & \vdots \\ \frac{\partial}{\partial x_d} g_1(x) & \frac{\partial}{\partial x_d} g_2(x) \end{bmatrix}$$

$\nabla g = \begin{bmatrix} \nabla g_1 & \nabla g_2 & \dots & \nabla g_k \end{bmatrix}^{d \times k}$

$\overset{d \times 1}{\nabla g_1} \rightarrow$ Gradient of the first co-ordinate $(x_1, x_2, \dots)$.

Let us look at convergence of each term on the right as $n \rightarrow \infty$. We will apply the multivariate version of continuous mapping theorem and Slutsky's theorem multiple times to our ingredient:

$$\sqrt{n}(\mathbf{T}_n - \vec{\theta}) \xrightarrow[n \rightarrow \infty]{(d)} \mathbf{T},$$

which also implies

$$(\mathbf{T}_n - \vec{\theta}) \xrightarrow[n \rightarrow \infty]{(d)/(p)} \mathbf{0}.$$

The first term $\nabla \mathbf{g}(\vec{\theta})^T (\sqrt{n}(\mathbf{T}_n - \vec{\theta}))$ is a continuous function of $(\sqrt{n}(\mathbf{T}_n - \vec{\theta}))$, hence

$$\nabla \mathbf{g}(\vec{\theta})^T (\sqrt{n}(\mathbf{T}_n - \vec{\theta})) \xrightarrow[n \rightarrow \infty]{(d)} (\nabla \mathbf{g}(\vec{\theta}))^T \mathbf{T} \text{ by continuous mapping theorem.}$$

For the second term, the first factor $\|\sqrt{n}(\mathbf{T}_n - \vec{\theta})\|$ is again a continuous function of $\sqrt{n}(\mathbf{T}_n - \vec{\theta})$, and therefore

$$\|\sqrt{n}(\mathbf{T}_n - \vec{\theta})\| \xrightarrow[n \rightarrow \infty]{(d)} \|\mathbf{T}\| \text{ by continuous mapping theorem.}$$

The second factor in the second term is a continuous function of $\mathbf{T}_n$ near $\vec{\theta}$. Hence

$$\mathbf{u}(\mathbf{T}_n) \xrightarrow[n \rightarrow \infty]{(d)/(p)} \mathbf{u}(\vec{\theta}) = \mathbf{0} \text{ by continuous mapping theorem.}$$

By (multivariate) Slutsky theorem, the entire second term converges to $\mathbf{0}$:

$$\|\sqrt{n}(\mathbf{T}_n - \vec{\theta})\| \mathbf{u}(\mathbf{T}_n) \xrightarrow[n \rightarrow \infty]{(d)/P} \|\mathbf{T}\| \mathbf{0} = \mathbf{0}.$$

Finally, applying the (multivariate) Slutsky theorem to the sum of the two terms gives:

$$\nabla \mathbf{g}(\vec{\theta})^T (\sqrt{n}(\mathbf{T}_n - \vec{\theta})) + \|\sqrt{n}(\mathbf{T}_n - \vec{\theta})\| \mathbf{u}(\mathbf{T}_n) \xrightarrow[n \rightarrow \infty]{(d)} \nabla \mathbf{g}(\vec{\theta})^T \mathbf{T} + \mathbf{0} = \nabla \mathbf{g}(\vec{\theta})^T \mathbf{T}.$$

LECTURE 10:

$$\Sigma = E[(X - \mu_X)(X - \mu_X)^T].$$

$$\Sigma = E[XX^T] - E[X]E[X]^T.$$

$$\Sigma = E[XX^T] - \mu_X \mu_X^T.$$

EXAMPLE:



Review of Covariance Matrices:

$$X = [x^{(1)}, x^{(2)}, \dots, x^{(d)}]^T.$$

$x_1, x_2, \dots$

$\frac{1}{2}, \frac{1}{2}$

$$E[X] = \frac{1}{2}.$$

$$\text{Var}(X) = \frac{1}{2} \left(1 - \frac{1}{2}\right).$$

$$= \frac{1}{4}.$$

d vectors $\rightarrow$ Probability $\rightarrow 1/d$.

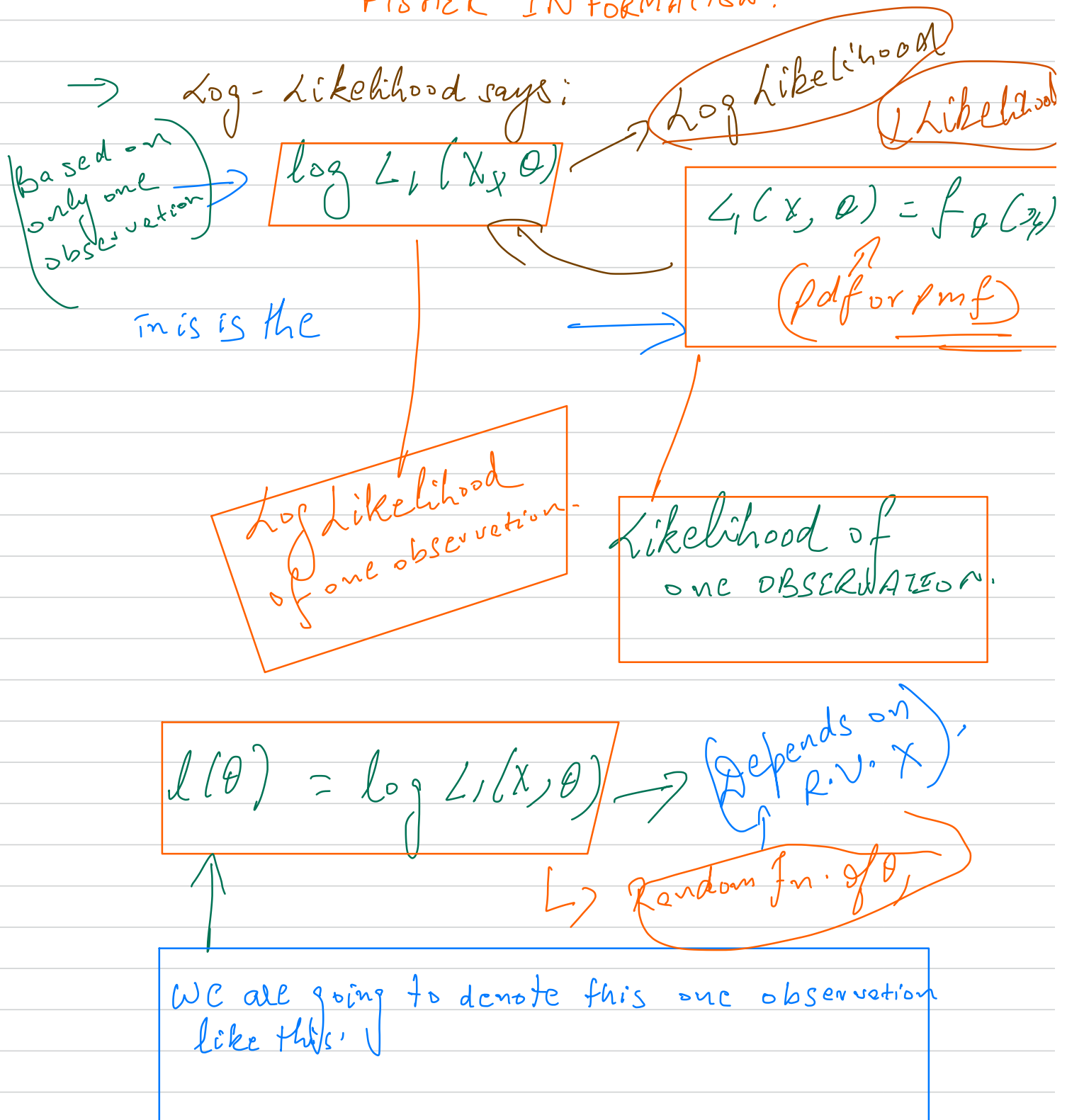
$$E[X_1] = x_1 \cdot 1/d = 1 \times 1/d = 1/d.$$

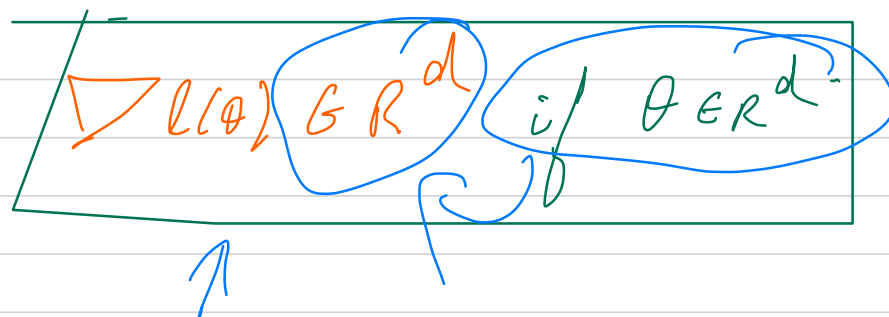
$$E[X_1^2] = x_1^2 (P(x)) = 1/d.$$

$$= x_1^2 P(x) = x_1 x_1^T P(x_1) = 1/d.$$

$$\therefore \text{Var}(x_1) = 1/d - 1/d^2 =$$

FISHER INFORMATION:





Random Vector

$$\text{Cov}(\nabla l(\theta)) = E[\nabla l(\theta) \nabla l(\theta)^T] - E[\nabla l(\theta)] E[\nabla l(\theta)]^T$$

Defⁿ: $\text{Cov}(\nabla l(\theta))$

$= -E[H l(\theta)]$

By Defⁿ of Fisher Information,

$$I(\theta) = \text{Cov}(\nabla l(\theta))$$

Hessian of $(H l(\theta))$
(2nd derivatives)

If $\theta \in \mathcal{R}_\theta$

$$I(\theta) = \text{Var}[l'(\theta)] = -E[l''(\theta)]$$

Sanity check → (After a few pages)

Exponential families:

→ A parametric family $\{P_\theta : \theta \in \Theta\}$

dominated by a σ -finite measure ν on $(\Omega, \mathcal{F})$.

if
$$\frac{dP_\theta}{d\nu}(\omega) = e^{\left[\eta(\theta)' T(\omega) - \psi(\theta) \right]}$$
$$h(\omega)$$
$$\omega \in \Omega$$

In simpler terms: (One Parameter if:).

$\eta(\theta), \psi(\theta)$ on Θ ,

T, h on $\mathbb{R}^q$,

$$p(x, \theta) = h(x) e^{\left[\eta(\theta)' T(x) - \psi(\theta) \right]}$$

$x \in \mathcal{X} \subset \mathbb{R}^q$; η, ψ and T
are not
unique.

($T(x)$ is sufficient for θ) 1-Parameter



We need to identify only

$e^{n(\theta)T(x) - b(\theta)}$ with $g(T(x), \theta)$ and $h(x)$



Factorisation Theorem.



with itself

States: $T(x)$ is sufficient for θ if there exist

$g(t, \theta), h(x)$

if $T(x) \propto t$
does not involve θ

such that

$$p(x, \theta) = g(T(x), \theta)h(x)$$

for $x \sim x_i$ (not iid) $\mathcal{P} = \{p_\theta : \theta \in \Theta\}$

$T(x) \rightarrow$ minimally sufficient for θ if

Any other sufficient statistic $S(x)$

s.t. ...

$$T(x) = \eta(S(x)).$$

Likelihood: X of θ , $L_X(\theta) = p(X, \theta)$, $\theta \in \Theta$.

$T(x) \rightarrow$ sufficient for θ

if $\theta_0 \in \Theta$.
s.t.

$$\{x : p(x, \theta) > 0\} \subset \{x : p(x, \theta_0) > 0\}$$

$\theta \in \Theta$.

By Factorisation Theorem,

Likelihood Ratio $\Lambda_X(\theta) = \frac{L_X(\theta)}{L_X(\theta_0)}$

Depends on x
through $T(x)$
only.

minimally
sufficient
statistic

$T \rightarrow$ natural sufficient
statistic

Poisson:

$$p(x, \theta) = \frac{\theta^x e^{-\theta}}{x!} = \frac{1}{x!}$$

$$e^{[x(\log \theta) - \theta]}$$

$\downarrow \quad \quad \downarrow \quad \quad \downarrow$
 $x \quad \quad \theta \quad \quad \theta$

$$\eta = 1, \quad \eta(\theta) = \log \theta$$

$$B(\theta) = \theta, \quad T(x) = x, \quad h(x) = \frac{1}{x!}$$

METHODS OF ESTIMATION:

→ How do we find $\theta(x)$ of $x \in \mathcal{X}$ close to unknown θ .

Consider: $P: (x) \times (\theta) \rightarrow \mathbb{R}$.

KL-divergence

$x \in \mathcal{X}, x \sim P, P \in \mathcal{P} = \{P_\theta : \theta \in \Theta\}$

Estimator.

Define: $D(\theta_0, \theta) = E_{\theta_0} p(x, \theta)$.

$D(\theta_0, \theta) \rightarrow$ UNIQUELY MINIMIZED
 $\forall \theta = \theta_0$

INFORMATION INEQUALITY:

Cramer's Lower Bound:

$x = (x_1, \dots, x_n)$ sample from $p \in \mathcal{P}$
 $= \{p_\theta : \theta \in \Theta\}$
 $\theta \in \mathbb{R}^k$.

Satisfies

$T(x)$ estimator

s.t.

$E[T(x)] = g(\theta) \rightarrow$ Differentiable on Θ .

(Lebesgue measure m on $(\mathbb{R}, \mathcal{B})$)
 $m([a, b]) = b - a$ ↳ Borel set

Probability measure μ ~~μ~~ $\mu \in ([0, 1], \mathcal{B}[0, 1])$
 ~~μ~~ $\mu \in ([0, 1], \mathcal{B}[0, 1])$

$$\frac{\partial}{\partial \theta} \int h(x) f_\theta(x) d\nu = \int h(x) \frac{\partial}{\partial \theta} f_\theta(x) d\nu$$

$\theta \in \Theta$
 $\downarrow$
in $\mathbb{R}^k$

if $h(x) \in \mathbb{R}^1$, $h(x) = T(x)$.

$$\text{Var}(T(X)) \geq \left[\frac{\partial g(\theta)}{\partial \theta} \right]^T [I(\theta)]^{-1} \frac{\partial g(\theta)}{\partial \theta}$$

$I(\theta) = E \left[\frac{\partial}{\partial \theta} \log f_{\theta}(X) \frac{\partial}{\partial \theta} \log f_{\theta}(X)^T \right]$
 (dxd) matrix. true semidefinite.

Proof: $k=1$

$$\text{Var}(T(X)) \geq \frac{[g'(\theta)]^2}{E \left[\frac{\partial}{\partial \theta} \log f_{\theta}(X) \right]^2}$$

$$\left[\text{Cov}(x_i, x_j) \right]^2 \leq \text{Var}(x_i) \text{Var}(x_j) \text{ if } i \neq j$$

$$\Rightarrow \left[E \left[\frac{\partial}{\partial \theta} \ln f_{\theta}(x) \right] \right]^2 = \text{Var} \left[\frac{\partial}{\partial \theta} \ln f_{\theta}(x) \right]$$

$$\Rightarrow g'(\theta) = \text{Cov} \left(T(x), \frac{\partial}{\partial \theta} \ln f_{\theta}(x) \right)$$

(Cramér's Lower Bound)

Theorem.

$$I(\theta) = -E[H(\theta)]$$

$$\text{Cov}(\nabla \theta) = E[\nabla \theta \nabla \theta^T] - E[\nabla \theta] E[\nabla \theta^T]$$

Variance

Log likelihood concave.

$$\left(\frac{\partial^2}{\partial \theta \partial \theta} \right) \leq 0$$
$$\therefore I(\theta) \geq 0$$

Imp.

When, $d=1$, $\theta \in \mathbb{R}$,

$$I(\theta) = \text{Var}(e^{\theta}) = -E[l''(\theta)]$$

fisher information

$I(\theta)$ can also be written as

$$I(\theta) = \int_{-\infty}^{\infty} \frac{\left(\frac{\partial}{\partial \theta} f(x) \right)^2}{f_{\theta}(x)} dx$$

$$I(\theta) = \text{Var}[\ell'(\theta)]$$

$$\text{Var}(x) = E[x^2] - E[x]^2$$

$$I(\theta) = - E[\ln L(\theta)]$$

$$I(\theta) =$$

For 1-d:

$$f_{\theta}(x) = L_1(x, \theta)$$

$$\int_{\mathcal{X}} f_{\theta}(x) dx = 1$$

$$\frac{\partial}{\partial \theta} \int_{\mathcal{X}} f_{\theta}(x) dx = 0$$

and also:

$$\frac{\partial^2}{\partial \theta^2} \int_{\mathcal{X}} L_1(x, \theta) dx = 0$$

$$l(\theta) = \sum \log L_1(x, \theta)$$

$$= \frac{\sum L_1(x, \theta)}{L_1(x, \theta)}$$

$$E[l'(\theta)] = \int \frac{\frac{\partial}{\partial \theta} L_1(x, \theta)}{L_1(x, \theta)} L_1(x, \theta) dx = 0 \quad \text{Density}$$

$$\Rightarrow \text{var}[l'(\theta)] = E[l'(\theta)^2]$$

Fisher Information

$$= \int \left(\frac{\frac{\partial}{\partial \theta} L_1(x, \theta)}{L_1(x, \theta)} \right)^2 L_1(x, \theta) dx$$

$$= \int \frac{\left(\frac{\partial}{\partial \theta} L_1(x, \theta) \right)^2}{L_1(x, \theta)} dx$$

$$-E[l''(\theta)] = - \int \frac{\frac{\partial^2}{\partial \theta^2} L_1(x, \theta) - \left(\frac{\partial}{\partial \theta} L_1(x, \theta) \right)^2}{L_1(x, \theta)^2} L_1(x, \theta) dx$$

$$= - \int_{\Theta^2} \mathcal{L}(x, \theta) dx + \text{Var}[\ell'(\theta)] = 0.$$

$$\Rightarrow -E[\ell''(\theta)] = \text{Var}[\ell'(\theta)].$$

↑
Fisher Information.

Definition of Fisher Information:

$$I(\theta) = \text{Cov}[\nabla \ell(\theta)] = -E[\mathcal{H}\ell(\theta)]$$

$$\ell(\theta) = \ln f_{\theta}(x)$$

Fisher Information of Bernoulli R.V.

If asked, compute Fisher Information of?

2 ways of doing it.

Compute 1st derivative.
↓
Take variance

Compute Hessian (2nd derivative)

Take expectation

Procedestimation does not pay

More difficult than taking an extra derivative

variance
↑
take derivative

$$X \sim \text{Ber}(p)$$

$$\rightarrow \ell(p)?$$

$$\text{pmf: } f_p(x) = p^x (1-p)^{1-x}$$

$$\ell(p) = x \log p + (1-x) \log (1-p)$$

$$\ell'(p) = \frac{x}{p} - \frac{1-x}{1-p}$$

$$\ell''(p) = -\frac{x}{p^2} - \frac{1-x}{(1-p)^2}$$

$$\begin{aligned} \text{Var}(\ell'(p)) &= \text{Var}\left(\frac{x}{p} - \frac{1-x}{1-p}\right) = \frac{1}{(1-p)^2} \\ &= \frac{p(1-p)}{p^4(1-p)^2} = \frac{1}{p(1-p)} \end{aligned}$$

$$E[\ell''(p)] = -\frac{E[x]}{p^2} - \frac{1-E[x]}{(1-p)^2} = -\frac{1}{p(1-p)}$$

$$\Rightarrow E[\ell''(p)] = \frac{1}{p(1-p)}$$

hello

Probability $\rightarrow$

Let $\{X_n\} \rightarrow X$.

$$\lim_{n \rightarrow \infty} P(|X_n - X| > \epsilon) = 0.$$

ie. $X_n \rightarrow X$.

DISTRIBUTION:

$$P(X_n \leq x) \xrightarrow{n \rightarrow \infty} P(X \leq x) = F(x).$$

$\hookrightarrow$ Continuous.

$\{X\}$, $X \rightarrow$ Random Variable

$X_n \xrightarrow{d} X$.

$x_n \xrightarrow{p} x$ then $x_n \xrightarrow{d} x$.
